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1  -- Count of direct reports that Grant Nguyen has
2
3  WITH DirectReports AS (
4      SELECT [EmployeeId], [First Name], [Last Name], [Manager]
5      FROM [Red30Tech].[dbo].[EmployeeDirectory$]
6      WHERE [EmployeeId] = 42
7      UNION ALL
8      SELECT e.[EmployeeId], e.[First Name], e.[Last Name], e.[Manager]
9      FROM [Red30Tech].[dbo].[EmployeeDirectory$] as e
10     INNER JOIN DirectReports as d ON e.[Manager] = d.[EmployeeId]
11 )
12
13 SELECT COUNT(*) as Direct_Reports
14 FROM DirectReports as d
15 WHERE d.[EmployeeId] != 42
16
17 -- DIRECT
18 SELECT count(*)
19 FROM [Red30Tech].[dbo].[EmployeeDirectory$] M
20 INNER JOIN [Red30Tech].[dbo].[EmployeeDirectory$] E
21 ON M.EmployeeId = E.Manager
22 WHERE M.EmployeeId= 42
23
24 ----- ↗
25     -----
26 -- Add CTE or Subquery for 3 orders with highest Order Totals from each ProdCategory purchased by Boehm Inc. ↗
27 WITH ROW_NUMBERS AS (
28     SELECT [OrderNum], [OrderDate], [CustName], [ProdCategory], [ProdName], [Order Total],
29     [Order Total],
30     ROW_NUMBER() OVER( PARTITION BY [ProdCategory] ORDER BY [Order Total]
31     DESC) as ROW_NUM
32 )
33 FROM [Red30Tech].[dbo].[OnlineRetailSales$]
34 WHERE [CustName] = 'Boehm Inc.'
35 )
36 SELECT * FROM ROW_NUMBERS WHERE ROW_NUM <= 3
37
38 ----- ↗
39     -----
40 LAG([Session Name], 1) OVER (Order by [Start Date] ASC) AS PreviousSession,
41 LEAD([Quantity_By_Day], 1) OVER (Order By [OrderDate] ASC) as
42 LastDateQuantity_1,
43
44 WITH ORDER_BY_DAYS AS (
45     SELECT [OrderDate], SUM([Quantity]) Quantity_By_Day
46     FROM [Red30Tech].[dbo].[OnlineRetailSales$]
47     WHERE [ProdCategory] = 'Drones'
48     GROUP BY ORDERDATE

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44 )
45 SELECT ORDERDate, Quantity_By_Day,
46 LAG([Quantity_By_Day], 1) OVER (Order By [OrderDate] ASC) as
    LastDateQuantity_1,
47 LAG([Quantity_By_Day], 2) OVER (Order By [OrderDate] ASC) as
    LastDateQuantity_2,
48 LAG([Quantity_By_Day], 3) OVER (Order By [OrderDate] ASC) as
    LastDateQuantity_3,
49 LAG([Quantity_By_Day], 4) OVER (Order By [OrderDate] ASC) as
    LastDateQuantity_4,
50 LAG([Quantity_By_Day], 5) OVER (Order By [OrderDate] ASC) as
    LastDateQuantity_5
51 FROM ORDER_BY_DAYS
52
53 -----
    -----
54 WITH RANKS AS (
55 SELECT *,
56 RANK() OVER (Partition BY STATE ORDER BY [Registration Date]) AS RANK_,
57 DENSE_RANK() OVER (Partition BY STATE ORDER BY [Registration Date]) AS
    DENSE_RANK_
58 FROM [Red30Tech].[dbo].[ConventionAttendees$]
59 )
60 SELECT * FROM RANKS
61
62 -----
    -----
63
64
65 -----
    -----
66
67
68 -----
    -----

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